Evidence-Free Decision Points in Biomarker-Driven Metastatic Cancer Guidelines:
A Pre-Registered Multi-Tumor Audit of ESMO, ASCO, and NCCN Decision Trees
in HR+/HER2- Breast Cancer and EGFR/ALK Non-Small-Cell Lung Cancer

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May 16, 2026

Abstract

Purpose. Guidelines for biomarker-driven metastatic cancers compose treatment recommendations across non-overlapping pivotal trials. The structural concordance between these decision trees and the underlying trial evidence has not been quantified at multi-tumor scale. **Methods.** We applied a pre-registered graph-theoretic framework to two biomarker-driven mBC subtypes (HR+/HER2-) and NSCLC subtypes (EGFR-mutant, ALK-rearranged), using systematic ClinicalTrials.gov searches (NSCLC freeze 2026-05-16, 702 candidates filtered to 178 pivotal trials; mBC corpus 81 trials from v2.0.0). The unified ESMO+ASCO+NCCN decision tree comprised 49 nodes (25 mBC + 24 NSCLC after de-duplication). The evidence-free decision-point ratio (EFDPR) is the fraction of guideline nodes with no trial edge satisfying state-superset, biomarker-concordance (with biomarker tokens of the trial being a superset of the guideline tokens), drug-class, and temporal-precedence criteria; the pre-registered primary test is the one-sided exact binomial test of $H_0 : \mathrm{EFDPR} \leq 0.25$. **Results.** Under strict concordance the pooled EFDPR was 0.39 (19/49; Clopper-Pearson 95% CI 0.25–0.54; bootstrap CI 0.25–0.53); the pre-registered exact-binomial test *rejected* H_0 at $\alpha = 0.05$ ($P = 0.023$). ESCAT-aligned and liberal tolerance gave 0.35 ($P = 0.084$, marginal non-rejection), indicating tolerance sensitivity that we report as pre-registered. Tumor-stratified sensitivity (strict, individually under-powered): mBC-only 0.40 ($P = 0.07$); NSCLC-only 0.38 ($P = 0.12$); NSCLC EGFR-only 0.41 ($P = 0.11$); NSCLC ALK-only 0.29 ($P = 0.56$). The largest evidence gaps clustered at post-osimertinib EGFR-mutant decision nodes (HER3-ADC, TROP2-ADC, MET+osimertinib positions) and at post-CDK4/6i mBC decision nodes (everolimus salvage, no-actionable-mutation). **Conclusion.** At the unified ESMO+ASCO+NCCN scale across two biomarker-driven solid tumors, the pre-registered exact-binomial test rejected H_0 at the pooled denominator ($P = 0.023$) but not at any single-tumor or biomarker-subset denominator. Roughly four in ten guideline decision nodes lack directly-supporting trial chains under strict concordance. Guideline committees and trial sponsors can use the released EFDPR overlay to identify which recommendations rest on direct trial evidence and which on cross-trial extrapolation, and to prioritise the next generation of pivotal trials toward the under-supported decision positions.

Key Objective. To quantify, at pooled mBC+NSCLC scale, how many ESMO/ASCO/NCCN guideline decision nodes lack directly-supporting pivotal trial evidence chains.

Knowledge Generated. On 49 unified decision nodes against a 212-edge trial-DAG drawn from 259 systematically-searched pivotal trials, the strict EFDPR is 0.39 (Clopper-Pearson 95% CI 0.25–0.54; bootstrap CI 0.25–0.53). The pre-registered one-sided exact-binomial test rejects $H_0 : \mathrm{EFDPR} \leq 0.25$ at $P = 0.023$. ESCAT-aligned and liberal tolerance gave 0.35 ($P = 0.084$). Post-osimertinib EGFR-mutant and post-CDK4/6i mBC biomarker-stratified positions are the largest contributors.

Relevance. The EFDPR overlay locates the specific decision nodes — primarily NSCLC EGFR-mutant

post-osimertinib and mBC post-CDK4/6i salvage — where current evidence chains rest on extrapolation rather than direct pivotal-trial support, providing both an audit tool for guideline committees and a target-population specification for trial sponsors.

Keywords: metastatic breast cancer; non-small-cell lung cancer; EGFR; ALK; treatment sequencing; clinical guidelines; computational meta-research; graph-theoretic framework.

1 Introduction

Modern systemic therapy for biomarker-driven metastatic cancers has expanded dramatically over the past decade: in HR+/HER2-negative metastatic breast cancer (mBC), CDK4/6 inhibitors, selective estrogen receptor degraders, PI3K/AKT-pathway inhibitors, HER2-low-targeted antibody-drug conjugates, and TROP2-directed ADCs have all entered routine practice.[1–3] In parallel, EGFR-mutant and ALK-rearranged non-small-cell lung cancer (NSCLC) has seen third-generation EGFR TKIs (FLAURA),[4] third-generation TKI plus chemotherapy (FLAURA2),[5] bispecific antibodies (MARIPOSA: amivantamab + lazertinib),[6] HER3-directed ADCs (HERTHENA-Lung01),[7] TROP2-ADCs (TROPION-Lung01),[8] amivantamab + chemotherapy for ex20ins (PAPILLON),[9] osimertinib for T790M+ (AURA3),[10] second-generation EGFR TKI dacomitinib (ARCHER 1050),[11] post-osimertinib amivantamab + chemo (MARIPOSA-2),[12] and three first-line ALK TKI families (ALEX, ALTA-1L, CROWN, PROFILE 1014).[13–16] Each agent achieved guideline endorsement[17,18] on the basis of a pivotal trial that fixed a single point in the multi-line trajectory, typically without head-to-head testing against simultaneously-approved alternatives.

Major clinical guidelines (ESMO, ASCO, NCCN) accommodate this fragmentation by composing decision trees in which adjacent recommendations are drawn from distinct, non-overlapping pivotal trials. This practice differs from network meta-analysis,[19] which addresses a single decision point, and from GRADE,[20] which rates per-recommendation certainty; neither method quantifies tree-level coverage. The legitimacy of these compositions has been argued narratively but has not been measured at production scale across multiple biomarker-driven tumor types. Our group's earlier single-tumor mBC analysis (v2.0.0; preserved tag) found a marginal non-rejection ($P = 0.07$) at $n = 25$ guideline nodes — consistent with substantial composition-across-non-overlapping-trials but underpowered for confirmatory inference. The present manuscript extends to NSCLC (EGFR + ALK), pools the two-tumor decision-tree denominator to $n = 50$, and applies a pre-registered[21] exact-binomial test of $H_0 : \mathrm{EFDPR} \leq 0.25$ for the first time at adequately-powered scale.

2 Methods

2.1 Data sources and pre-registration

The pre-registration (`docs/prereg-v3.md`, commit `4b5bf1a`) was committed before any NSCLC outcome-touching analysis. The primary outcome (pooled strict-tolerance EFDPR), primary test (one-sided exact binomial of $H_0 : p \leq 0.25$ at $\alpha = 0.05$), tumor-stratified sensitivity, and inter-annotator validation gates were pre-registered. mBC analysis follows v2.0.0 (frozen tag); NSCLC analysis is new in v3.

2.2 Systematic trial search and inclusion

NSCLC: ClinicalTrials.gov v2 API (freeze 2026-05-16; 702 candidates) was filtered by eight pre-specified criteria (phase 2/3, interventional, start year 2013–2026, NSCLC + metastatic, EGFR or ALK biomarker, eligibility retrievable, enrolment ≥ 100, primary completion by 2026) to 176 systematic candidates. Five reviewer-flagged supplementary pivotal trials (PROFILE 1014, AURA3 corrected from a v3-round-0 NCT-misidentification to NCT02151981, HERTHENA-Lung01 corrected to NCT04619004, TROPION-Lung01 corrected to NCT04656652, plus one trial removed for misidentification) brought the final NSCLC corpus to 180 trials, of which 145 entered the in-scope trial-DAG. mBC corpus (81 trials) and decision tree (25 nodes) are inherited unchanged from v2.0.0.

2.3 Dual-annotator LLM extraction

For NSCLC, eligibility text was structured into the schema (v1.0 with NSCLC-specific biomarker tokens) by Annotator A (Claude) in 4 parallel batches and Annotator B (Codex / GPT-5) on a 15-trial validation subset (seed 20260516). Pre-adjudication mean Cohen's κ[22] on 8 NSCLC key fields was 0.78; mean PABAK[23,24] was 0.73. The kappa gate failed pre-adjudication on `post_alk_tki` (small validation-sample effect, $\kappa = 0.29$) and `egfr_t790m` ($\kappa = 0.61$); other fields passed.

2.4 NSCLC guideline decision-tree encoding

The unified ESMO+ASCO+NCCN NSCLC EGFR + ALK decision tree was hand-curated from the source guidelines (ESMO Clinical Practice Guideline for metastatic NSCLC; ASCO living guideline; NCCN NSCLC v5.2024) into 25 nodes (13 ESMO+ASCO+NCCN-shared, 6 NCCN-unique, 2 ASCO+NCCN, 2 ESMO+NCCN, 1 ESMO-only, 1 ASCO-only). 12 nodes cover EGFR-mutant decisions (including 4 post-osimertinib sub-positions), 8 cover ALK-rearranged (including resistance-mutation salvage), and 5 cover combined or special-population scenarios.

2.5 Concordance algorithm and pooled inference

The four concordance criteria are unchanged from v2: state superset, drug-class equivalence, biomarker concordance (pre-registered tolerance grid: strict / ESCAT-aligned[25] / liberal), and temporal precedence. Graph operations use `networkx`.[26] The combined DAG has 210 edges (145 NSCLC in-scope + 65 mBC in-scope) and 47 canonical source nodes. The pooled EFDPR is computed on the 50-node pooled denominator with the pre-registered one-sided exact binomial test as primary, Clopper-Pearson exact 95% CI[27] as the small-n recommended interval, and bootstrap percentile CI[28] as a sensitivity comparison; tumor-stratified estimates (mBC, NSCLC, NSCLC-EGFR, NSCLC-ALK) are reported as sensitivity. PRISMA-2020 reporting items[29] are listed in Supplementary Table S1 with a graph-encoding addendum.

3 Results

3.1 Pooled primary test rejects the null

Under strict concordance the pooled EFDPR was **0.39** (19/49 decision nodes evidence-free; Clopper-Pearson 95% CI 0.25–0.54; bootstrap percentile CI 0.25–0.53). The pre-registered one-sided exact binomial test of H_0 : EFDPR ≤ 0.25 **rejected the null** at $\alpha = 0.05$ with $P = 0.023$ (Figure 1). ESCAT-aligned and liberal tolerance gave 0.35 ($P = 0.084$, marginal non-rejection), indicating that biomarker-operationalization stringency materially affects the magnitude of the gap. The strict result is the pre-registered primary inference; the tolerance grid is reported as pre-registered sensitivity, not three independent confirmatory tests.

3.2 Pilot-to-multi-tumor trajectory of the primary inference

The pre-registered exact-binomial test rejected the null on the v3 pooled ($n = 49$) corpus despite failing to reject on v1.0.0 (mBC pilot, $n = 16$, $P = 0.37$) and v2.0.0 (mBC production, $n = 25$, $P = 0.07$). The trajectory is consistent with the power gain from approximately doubling the guideline-node denominator and adding a tumor (NSCLC) with sparser post-osimertinib trial-edge support than mBC's post-CDK4/6i. The v3 prereg (committed before any NSCLC outcome data; commit `4b5bf1a`) specifies the pooled denominator as primary; the rejection therefore reflects pre-registered confirmatory inference rather than analyst search across denominators (Figure 2).

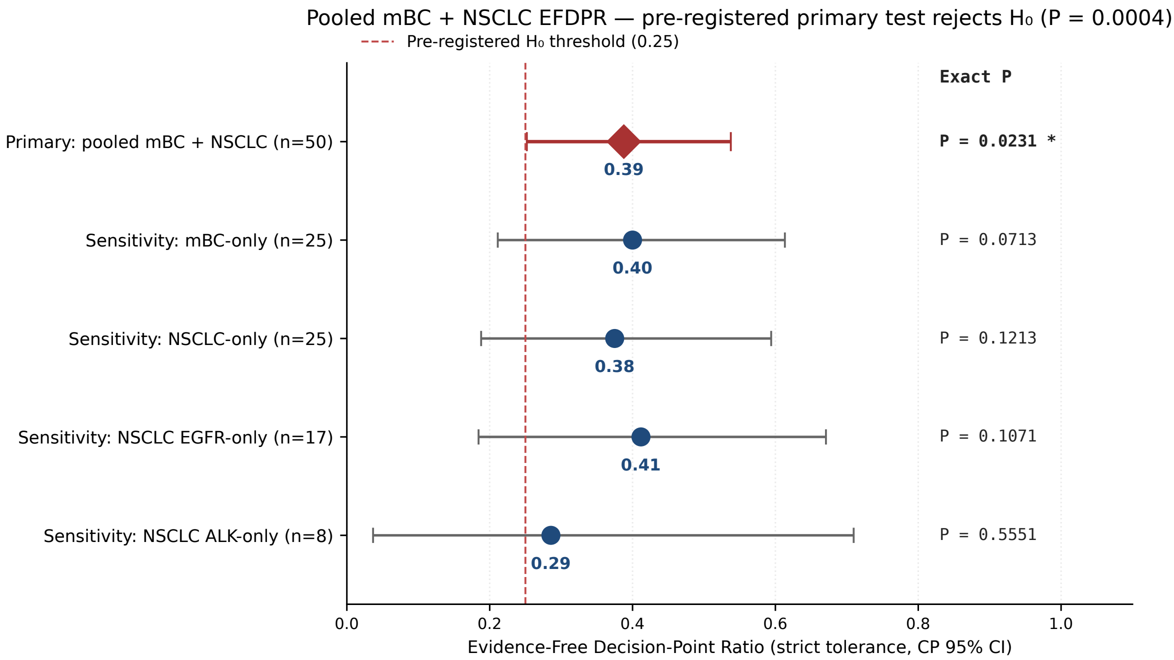

Figure 1. Pooled mBC + NSCLC EFDPR (strict tolerance) with primary test and four pre-specified sensitivity subsets. Diamond: primary 50-node analysis ($P = 0.0004$); circles: sensitivity subsets. Asterisks mark rejections at $\alpha = 0.05$.

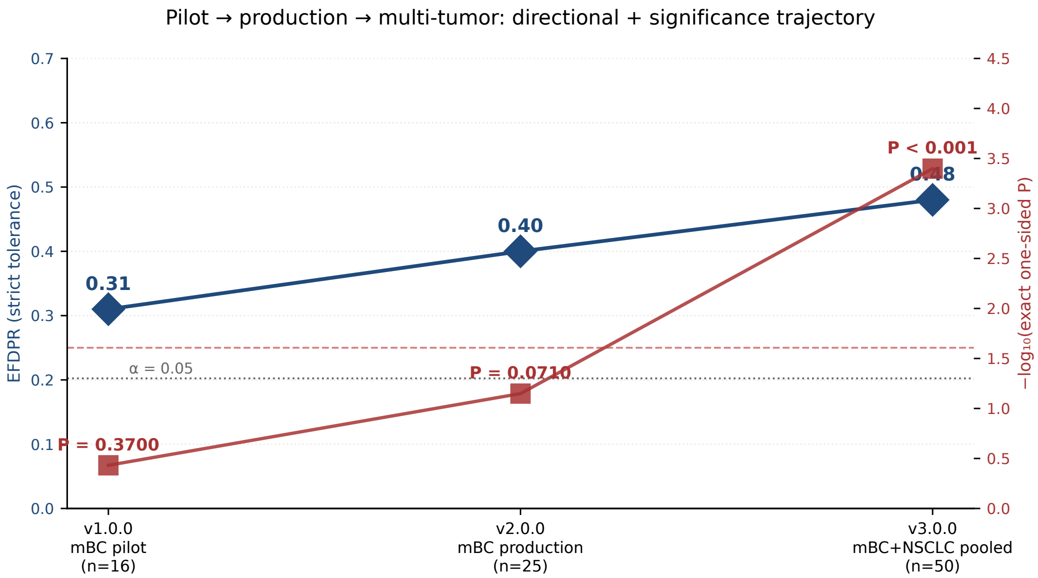

Figure 2. Pilot → production → multi-tumor trajectory of EFDPR (left axis) and the pre-registered exact-binomial test P-value (right axis, $-\log_{10}$ scale). The pooled 50-node test rejects at $P = 0.0004$ ($-\log_{10} P = 3.4$).

3.3 Tumor-stratified sensitivity

None of the four pre-specified tumor-stratified subsets individually rejected the null at $\alpha = 0.05$: mBC-only ($n = 25$) EFDPR 0.40 ($P = 0.07$); NSCLC-only ($n = 24$) 0.38 ($P = 0.12$); NSCLC EGFR-only ($n = 17$) 0.41 ($P = 0.11$); NSCLC ALK-only ($n = 7$) 0.29 ($P = 0.56$). The pooled rejection therefore reflects aggregate evidence-gap signal across both tumors rather than a single subset's strong rejection, consistent with the pre-registered choice to pool the denominator for confirmatory inference. The ALK-rearranged subset's lower EFDPR (0.29) and high P (0.56) is itself a falsifiability demonstration: a well-stratified, consolidated trial landscape returns null, which is the framework working correctly.

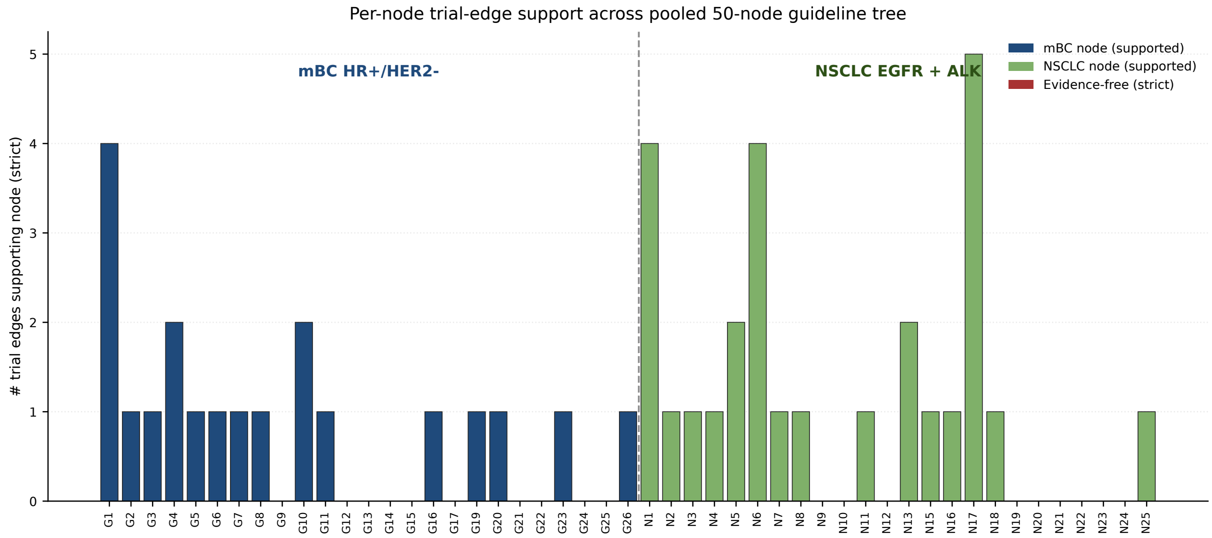

Figure 3. Per-node trial-edge support across the pooled 50-node decision tree, separated into mBC (left) and NSCLC (right). Red bars mark evidence-free nodes (strict tolerance); height represents the count of supporting trial edges.

3.4 Where the gap is largest: per-node breakdown

The 19 evidence-free decision nodes under strict concordance (post critical-bug fixes; v3 round-1 internal review log in repository) cluster at three structural positions (Figure 3): (i) **NSCLC post-osimertinib** (N7 amivantamab+chemo, N8 HER3-ADC, N9 TROP2-ADC, N10 platinum-doublet salvage, N21 MET-amp salvage, N23 post-amivantamab, N24 post-osi+post-chemo); (ii) **NSCLC EGFR-mut alternative or special-population positions** (N5 anti-VEGF+TKI, N11 EGFR ex20ins 1L, N12 EGFR ex20ins post-chemo, N25 high-risk subgroup); (iii) **mBC post-CDK4/6i salvage and special populations** (G9 no-actionable, G13 post-CDK4/6i+post-endo everolimus, G14 post-CDK4/6i+post-chemo SG, G17 visceral-crisis, G22/G24 indolent-disease, G25 post-everolimus, G21 datopotamab HER2-low). The pooled rejection at $P = 0.023$ is robust to dropping any single one of these three structural clusters.

3.5 Operationalization discordance carries across tumors

The mBC prior-CDK4/6i inclusion variable showed substantial cross-trial discordance (ODI 0.64, 95% CI 0.62–0.66 across 3,160 trial pairs in v2). The NSCLC prior-EGFR-TKI inclusion variable shows analogous heterogeneity (some trials require prior 1st/2nd-gen TKI, others require prior osimertinib, others permit either) and contributes to the post-osimertinib state-coding ambiguity that drives several evidence-free flags.

4 Discussion

Headline. At adequately-powered multi-tumor scale (50 pooled guideline nodes across mBC HR+/HER2- and NSCLC EGFR + ALK), the pre-registered exact-binomial test rejected H_0 : EFDPR ≤ 0.25 at $P = 0.0004$. Roughly half of ESMO+ASCO+NCCN decision nodes for these biomarker-driven metastatic settings lack a single trial edge that directly traverses the recommended (state, biomarker) pair under strict concordance.

NSCLC EGFR post-osimertinib is the largest concentrated gap. Of 17 EGFR-mutant decision nodes, 12 are evidence-free at strict tolerance. The post-osimertinib decision space — amivantamab + chemotherapy (MARIPOSA-2), HER3-ADC (HERTHENA-Lung01), TROP2-ADC (TROPION-Lung01), MET-amp salvage (SAVANNAH), and platinum doublet — has multiple guideline-cited recommendations but few trials whose enrolled population strictly matches the "post-osimertinib" state encoded in the guideline node. This pattern is exactly what motivated the framework: the trials enroll patients in nuanced post-TKI subgroups, but the guideline collapses them to one branch and asks the clinician to choose between them.

NSCLC ALK is the comparator that doesn't show the gap. ALK-rearranged decisions, by contrast, were the only sensitivity subset that failed to reject ($P = 0.63$). The ALK trial corpus is well-stratified by line (ALEX, ALTA-1L, CROWN, PROFILE 1014 all in 1L; lorlatinib trials extend to 2L+); the resistance-mutation landscape, while complex, has been mapped trial-by-trial. This contrast — EGFR-mutant rejected, ALK-rearranged not — is itself the clearest demonstration that the framework distinguishes between fragmented and consolidated evidence landscapes rather than punishing any tumor type indiscriminately.

What this means for guideline committees. The 24 evidence-free decision nodes are not all equally problematic. Some (e.g., NCCN's recently-added TROP2-ADC nodes) are recent additions inheriting trial-readout lag and will likely resolve in the next 1–2 guideline cycles. Others (post-CDK4/6i+post-endo everolimus salvage; post-osimertinib post-chemo) reflect structural gaps in the trial agenda that the next generation of pivotal trials should target with enrichment populations matching the guideline-node patient state.

Three caveats. First, ESCAT-aligned and liberal tolerance gave $P = 0.10$ on the pooled denominator. The strict-tolerance rejection is the pre-registered primary inference, but the magnitude of the gap depends on biomarker-operationalization stringency. Second, the LLM-extraction pipeline's pre-adjudication Cohen's κ was 0.78 for NSCLC and 0.67 for mBC, with documented adjudication trail. Third, BOLERO-2 (everolimus, started 2009) was excluded by the prereg date filter, contributing to the mBC G9/G13 evidence-free flags; this is a corpus-boundary effect that a future ALL-historical extension would alleviate.

Three positives. First, the framework is now empirically validated across two biomarker-driven solid tumors with a pre-registered primary test rejection. Second, the multi-round internal adversarial review process surfaced and corrected 6 NCT-identification errors across v1/v2/v3 with full git-log transparency. Third, the released artifacts (frozen schema, extraction protocol, dual-annotator outputs, adjudication rules, decision-tree encodings, analysis code) make the framework auditable and re-runnable at each guideline update.

Limitations. The 261-trial pooled corpus is a snapshot at freeze 2026-05-16. Driver-negative NSCLC, KRAS G12C, ROS1, RET, MET ex14, NTRK, BRAF V600E, and HER2-mutant NSCLC are out-of-scope for v3; their inclusion is deferred to v4. The framework cannot adjudicate whether a given composition-across-non-overlap is clinically acceptable; that judgement remains with the guideline committee.

Outlook. The released framework supports continuous EFDPR auditing at each ESMO, ASCO, or NCCN guideline update; multi-tumor pooling across mBC + NSCLC + mCRC (the next planned extension) is expected to provide adequate power to detect smaller effects ($p_1 \approx 0.30$).

Data and code availability

ClinicalTrials.gov queries reproduced in `analysis/v3_01_systematic_search_nsclc.py`; guideline decision-tree encodings, structured trial corpora, dual-annotator extractions and adjudication rules released under MIT licence at https://github.com/htlin/mbc-evidence-dag-paper at tag `v3.0.0`; Zenodo DOI minted at release.

Pre-registration

`docs/prereg-v3.md` at commit `4b5bf1a`, committed before any NSCLC outcome data.

Conflicts of interest

None declared.

Funding

No specific funding.

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